# Does Removing Non-Anatomical Artifacts from Chest X-Rays Improve Pneumonia Classification?

Aleksandr Sorokin Viggo Beck Sebastian Drumm
Hero Engelund

Applied Artificial Intelligence — Exam Project

June 4, 2026

## Abstract

Chest X-rays carry burned-in orientation markers (the letters $R/L$) and other non-anatomical text that a convolutional neural network (CNN) might exploit as a shortcut instead of learning genuine pathology. We ask whether removing these artifacts improves pneumonia classification. Following Nunamaker et al.'s multi-methodological process, we built a reproducible PyTorch pipeline on the Kaggle *Chest X-Ray Images (Pneumonia)* dataset (5,856 images), re-split 80/10/10, and trained a custom three-block CNN. We then ran a *locked* experiment: one model trained on the original images and one on a copy in which bright marker artifacts were detected with OpenCV and inpainted, with every other factor held identical. The cleaned dataset produced essentially no change in held-out test performance (Youden's J $-0.0023$, AUC $-0.0001$, accuracy $-0.0017$). We conclude that, for this dataset and architecture, removing the markers neither helps nor hurts: the classifier's decisions are already driven by the lung fields rather than the markers.

## 1 Introduction

Pneumonia is a leading cause of death worldwide, and the chest X-ray remains the most common first-line imaging examination for diagnosing it. Deep convolutional neural networks now match or exceed clinicians on several chest-X-ray tasks [1], which makes them attractive as triage tools.

A recurring danger in medical imaging, however, is *shortcut learning*: a model reaches high accuracy by latching onto spurious correlations rather than the intended signal [2]. Chest radiographs are a textbook case. Zech et al. showed that CNNs can key on hospital-specific tokens burned into the image and fail to generalise across institutions [3], and DeGrave et al. found that COVID-19 X-ray classifiers frequently relied on laterality markers, text and image edges instead of lung pathology [4]. Our dataset contains exactly such artifacts: most films carry a bright $R$ (or $L$) orientation marker and occasional technician text near the border (Figure 3).

This motivates a concrete, testable question: if these non-anatomical artifacts were removed, would a pneumonia classifier perform better? A positive answer would suggest the model is partly cheating; a negative answer would provide evidence that the markers are not driving its decisions. We investigate this with a controlled, reproducible two-dataset experiment.

## 2 Research Question

We phrase one overall question with two supporting subquestions.

**Q1** Does removing non-anatomical artifacts (e.g. the orientation marker $R$) from chest X-rays improve pneumonia classification performance?

**Q1.1** How can the artifacts be detected and removed reliably and time-efficiently, without damaging anatomy?

**Q1.2** How should the strong class imbalance between NORMAL and PNEUMONIA images be handled so that it does not confound the comparison?

To answer Q1 we keep *every* factor constant between two training runs and change only one thing — whether the input images still contain the artifacts — so that any difference in test performance is attributable to the cleaning alone.

## 3 Methods

We follow Nunamaker et al.'s multi-methodological systems-development process [5], iterating between observation, systems development and experimentation. Every step from data import to evaluation is scripted so the results are transparent and reproducible; all random number generators (Python, NumPy and PyTorch, CPU and CUDA) are seeded with a fixed value.

### 3.1 Data and splitting

We use the Kaggle *Chest X-Ray Images (Pneumonia)* dataset [6, 7], comprising 5,856 paediatric frontal chest radiographs labelled NORMAL or PNEUMONIA. The original Kaggle partition has a very small validation set (16 images), so we merge all images and re-split them *by file* into 80 %/10 %/10 % train/validation/test partitions with a fixed seed. The same split membership is reused for both the original and the cleaned dataset so the comparison is image-for-image identical (Section 3.6).

### 3.2 Preprocessing

Images are converted to single-channel grayscale, padded to a square (to avoid distortion), and resized to $224 \times 224$ px — the standard input size for this family of models [8]. Pixel values are normalised to a mean and standard deviation of 0.5. During training only, we apply light augmentation (rotation $\pm 7°$, translation $\pm 3\,\%$, zoom $\pm 3\,\%$); augmentation is a standard regulariser that improves generalisation on limited medical data [9]. Validation and test images receive no augmentation.

### 3.3 Class imbalance

The dataset is imbalanced (73 % PNEUMONIA, 27 % NORMAL), which can bias a classifier toward the majority class [10, 11]. We address this with a `WeightedRandomSampler` that draws minority-class images more often, so each training batch is approximately class-balanced. The sampler is applied only to the *training* loader; validation and test sets keep their natural distribution so reported metrics reflect realistic prevalence.

### 3.4 Model architecture

We use a custom three-block CNN (Figure 1). Each block is a $3 \times 3$ convolution followed by batch normalisation [12], a ReLU activation and $2\times2$ max-pooling; the channel width grows $1 \to 32 \to 64 \to 128$. The resulting $128 \times 28 \times 28$ feature map is flattened into a fully connected classifier with a 256-unit hidden layer, dropout [13] and a two-way output. The design follows the classic convolution–pooling–dense template established for image classification [14, 15] ($\approx$25.8 M parameters).

### 3.5 Artifact detection and removal

Orientation markers and technician text are bright, small, near-white blobs that sit on a dark background close to the image border, away from the lung fields. We detect them with OpenCV [16] using three independent tests on the connected components of the brightest pixels: *(i)* location — the blob's centroid lies in a left, right or top edge band (the bottom is excluded, where the diaphragm and film strips appear); *(ii)* shape — a letter-like size, aspect ratio and solidity; and *(iii)* isolation — a dark ring of surrounding pixels, which distinguishes a marker from bright anatomy (ribs, clavicles) that is connected to other bright tissue. Detected regions are dilated and filled with Telea inpainting [17], which reconstructs the masked area from neighbouring pixels so no hard edge is left behind. The procedure is scripted (`create_cleaned_dataset.py`) to write a second dataset with an identical folder structure.

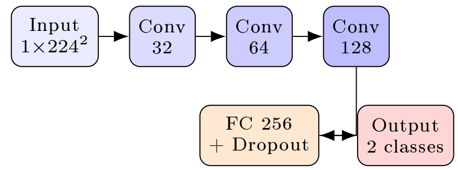


**Figure 1:** Three-block CNN. Each Conv block = 3×3 convolution + batch norm + ReLU + 2×2 max-pool; the flattened features feed a dropout classifier.

### 3.6 Locked two-dataset experiment

To answer Q1 we train two fresh models — one on the original images, one on the cleaned images — while *locking* the architecture, hyperparameters, split membership, preprocessing, sampler logic, batch size and threshold-selection method. The only variable is the dataset path, so any performance difference is caused by artifact removal alone.

### 3.7 Hyperparameter search

Before the locked comparison we tune the baseline with a 20-trial random search [18] over learning rate (log-uniform $[10^{-5}, 10^{-3}]$), dropout ($[0.2, 0.6]$), weight decay (log-uniform $[10^{-5}, 10^{-2}]$) and batch size $\{16, 32, 64\}$, optimising with AdamW and a `ReduceLROnPlateau` schedule. Trial 1 is the default configuration so the search directly answers "did tuning beat the default?".

### 3.8 Evaluation

Because missing a true pneumonia case is costly, accuracy alone is inadequate; we report sensitivity (recall), specificity, precision, the area under the ROC curve (AUC), and Youden's J statistic $J = \text{sensitivity} + \text{specificity} - 1$ [19]. The decision threshold for each model is selected on the *validation* set (the point that maximises $J$) and then applied once to the held-out test set, which is touched only for the final evaluation.

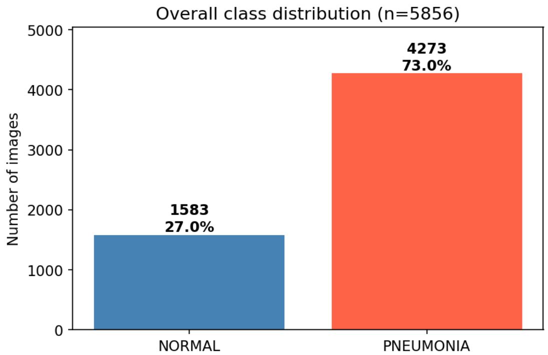


**Figure 2:** Overall class distribution across the merged dataset. PNEUMONIA images outnumber NORMAL images roughly 2.7:1.

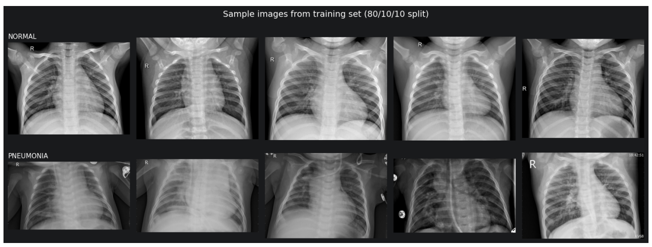


**Figure 3:** Sample training images (top: NORMAL, bottom: PNEUMONIA). Bright orientation markers are visible near the borders.

# 4 Analysis

## 4.1 Data observation

The re-split data contains 4,684 training, 585 validation and 587 test images. The class imbalance is clear (Figure 2), confirming the need for the weighted sampler of Section 3.3. Sample images (Figure 3) reveal the orientation markers and confirm the visual difference between the classes.

## 4.2 Artifact removal

Applying the detector of Section 3.5 localises the bright border markers and inpaints them cleanly while leaving the lung fields untouched (Figure 4). Visual inspection across many images confirms that the three-test rule rarely fires on anatomy, satisfying sub-question Q1.1.

## 4.3 Hyperparameter search

The random search (Section 3.7) explored 20 configurations. The best trial improved validation Youden's J over the default by +0.021 (Table 1), confirming the default configuration was already near a good operating point; the tuned and default models are therefore both reasonable baselines for the locked comparison.

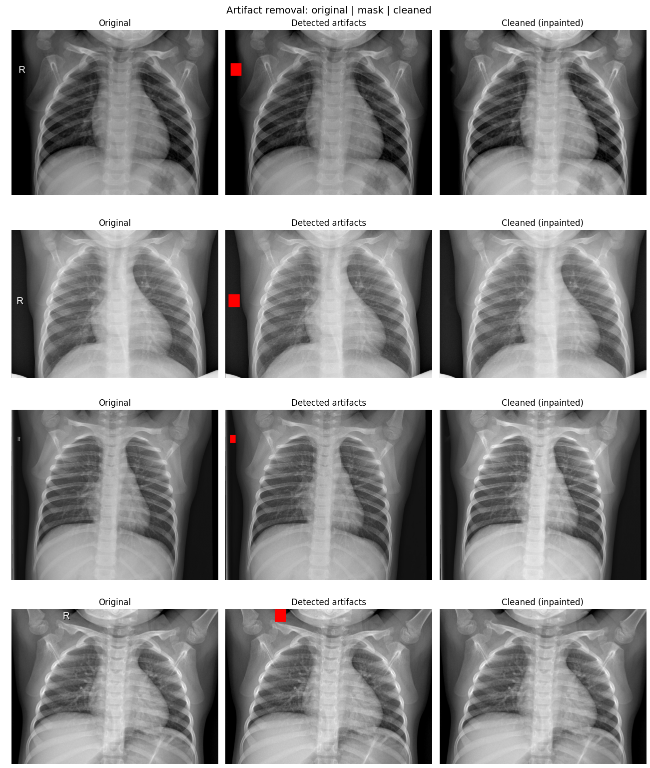


**Figure 4:** Artifact removal. Left: original. Middle: detected marker region (red). Right: image after Telea inpainting. The $R$ markers are removed without disturbing the lungs.

**Table 1:** Top random-search trials, ranked by validation Youden's J (the default configuration is included for reference).

| Trial | lr | dropout | wd | Val $J$ | Val AUC |
|---|---|---|---|---|---|
| random-3 | 1.5e-5 | 0.37 | 1.2e-5 | **0.9065** | 0.9851 |
| random-1 | 1.9e-4 | 0.21 | 6.7e-5 | 0.9002 | 0.9870 |
| random-5 | 2.0e-4 | 0.42 | 4.6e-5 | 0.9002 | 0.9826 |
| default | 1.0e-4 | 0.40 | 0.0 | 0.8852 | 0.9857 |

## 4.4 Original vs. cleaned

We then ran the locked experiment of Section 3.6. Both models were trained from scratch under identical settings and evaluated on the same held-out test set with validation-selected thresholds. The test confusion matrices (Figure 5) and the metric comparison (Figure 6, Table 2) are reported in Section 5.

# 5 Findings

On the held-out test set the original-data model achieved a Youden's J of 0.8226 (sensitivity 0.8855, specificity 0.9371, accuracy 0.8995, AUC 0.9734). The model trained on the cleaned data achieved a Youden's J of 0.8203 (sensitivity 0.8832, specificity 0.9371, accuracy

**Table 2:** Held-out test performance for the locked two-dataset experiment. $\Delta$ is cleaned minus original.

| Metric | Original | Cleaned | $\Delta$ |
|---|---|---|---|
| Youden's J | 0.8226 | 0.8203 | −0.0023 |
| Sensitivity | 0.8855 | 0.8832 | −0.0023 |
| Specificity | 0.9371 | 0.9371 | 0.0000 |
| Accuracy | 0.8995 | 0.8978 | −0.0017 |
| AUC | 0.9734 | 0.9733 | −0.0001 |

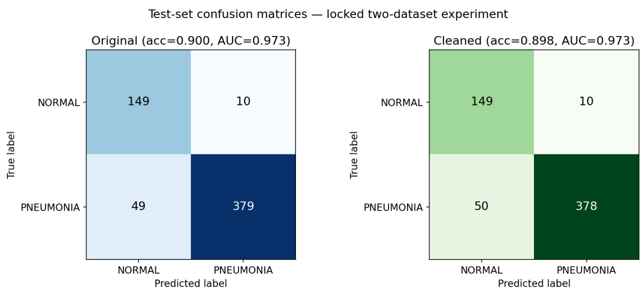


**Figure 5:** Test-set confusion matrices for the model trained on the original images (left) and on the cleaned images (right).

0.8978, AUC 0.9733). The differences (cleaned minus original) are −0.0023 in Youden's J, −0.0023 in sensitivity, 0.0000 in specificity, −0.0017 in accuracy and −0.0001 in AUC (Table 2).

The confusion matrices (Figure 5) differ by a single case: the cleaned model classifies one additional PNEUMONIA image as NORMAL (a false negative). All other cells are identical. The ROC-level ranking, summarised by AUC, is unchanged to three decimal places.

# 6 Conclusion

Returning to **Q1** (Section 2): for this dataset and architecture, removing non-anatomical artifacts did *not* improve pneumonia classification. Every test metric moved by less than 0.25 percentage points and all changes were slightly negative (Table 2, Section 5) — well within the run-to-run noise we would expect from a single split. We therefore find no evidence that the baseline model was relying on the orientation markers; its decisions appear to be driven by the lung fields, not the burned-in text. In the language of Section 1, the feared shortcut [2, 4] does not materially affect performance here, most likely because the weighted sampler, augmentation and the markers' position outside the lungs already limit their influence.

On the supporting questions, the OpenCV-plus-inpainting detector (Q1.1) removed markers reliably and cheaply without harming anatomy (Figure 4), and the weighted sampler (Q1.2) kept the comparison fair by neutralising the class imbalance.

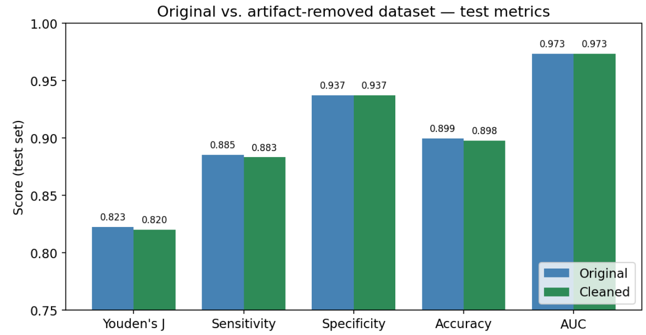


**Figure 6:** Test metrics for the original and cleaned datasets. The bars are visually indistinguishable across all five metrics.

These conclusions carry clear limitations. We tested a single custom CNN on a single re-split of one paediatric dataset, and our cleaning targets only bright *border* markers rather than performing full lung-field segmentation, so residual non-anatomical content may remain. A negative result on one split is also not proof of no effect. Future work should (i) repeat the locked experiment across several seeds and report confidence intervals, (ii) replace marker removal with a pretrained U-Net lung segmentation so the model sees only lung tissue, and (iii) use Grad-CAM saliency maps to verify *directly* where the network looks. Such explainability evidence would complement the behavioural result reported here.

# References

[1] P. Rajpurkar et al., "CheXNet: Radiologist-Level Pneumonia Detection on Chest X-Rays with Deep Learning," *arXiv:1711.05225*, 2017.

[2] R. Geirhos et al., "Shortcut Learning in Deep Neural Networks," *Nature Machine Intelligence*, vol. 2, pp. 665–673, 2020.

[3] J. R. Zech et al., "Variable generalization performance of a deep learning model to detect pneumonia in chest radiographs: A cross-sectional study," *PLOS Medicine*, vol. 15, no. 11, e1002683, 2018.

[4] A. J. DeGrave, J. D. Janizek, S.-I. Lee, "AI for radiographic COVID-19 detection selects shortcuts over signal," *Nature Machine Intelligence*, vol. 3, pp. 610–619, 2021.

[5] J. F. Nunamaker Jr., M. Chen, T. D. M. Purdin, "Systems Development in Information Systems Research," *Journal of Management Information Systems*, vol. 7, no. 3, pp. 89–106, 1990.

[6] D. S. Kermany et al., "Identifying Medical Diagnoses and Treatable Diseases by Image-Based Deep Learning," *Cell*, vol. 172, no. 5, pp. 1122–1131, 2018.

[7] P. Mooney, "Chest X-Ray Images (Pneumonia)," Kaggle dataset, 2018. https://www.kaggle.com/datasets/paultimothymooney/chest-xray-pneumonia.

[8] K. He, X. Zhang, S. Ren, J. Sun, "Deep Residual Learning for Image Recognition," *CVPR*, 2016.

[9] C. Shorten, T. M. Khoshgoftaar, "A Survey on Image Data Augmentation for Deep Learning," *Journal of Big Data*, vol. 6, no. 60, 2019.

[10] H. He, E. A. Garcia, "Learning from Imbalanced Data," *IEEE Transactions on Knowledge and Data Engineering*, vol. 21, no. 9, pp. 1263–1284, 2009.

[11] B. Krawczyk, "Learning from imbalanced data: open challenges and future directions," *Progress in Artificial Intelligence*, vol. 5, pp. 221–232, 2016.

[12] S. Ioffe, C. Szegedy, "Batch Normalization: Accelerating Deep Network Training by Reducing Internal Covariate Shift," *ICML*, 2015.

[13] N. Srivastava et al., "Dropout: A Simple Way to Prevent Neural Networks from Overfitting," *Journal of Machine Learning Research*, vol. 15, pp. 1929–1958, 2014.

[14] Y. LeCun, L. Bottou, Y. Bengio, P. Haffner, "Gradient-Based Learning Applied to Document Recognition," *Proceedings of the IEEE*, vol. 86, no. 11, pp. 2278–2324, 1998.

[15] A. Krizhevsky, I. Sutskever, G. E. Hinton, "ImageNet Classification with Deep Convolutional Neural Networks," *NeurIPS*, 2012.

[16] G. Bradski, "The OpenCV Library," *Dr. Dobb's Journal of Software Tools*, 2000.

[17] A. Telea, "An Image Inpainting Technique Based on the Fast Marching Method," *Journal of Graphics Tools*, vol. 9, no. 1, pp. 23–34, 2004.

[18] J. Bergstra, Y. Bengio, "Random Search for Hyper-Parameter Optimization," *Journal of Machine Learning Research*, vol. 13, pp. 281–305, 2012.

[19] W. J. Youden, "Index for rating diagnostic tests," *Cancer*, vol. 3, no. 1, pp. 32–35, 1950.